True parameter (true state)

$$\theta_t = \binom{\alpha_t}{\beta_t}$$

$$\theta_t = \theta_{t-1} + \omega_t \quad \omega_t \sim N(0, Q)$$

Observation

$$F_t = (1 \quad X_t)$$

$$\mathrm{Y}_t = F_t\theta_t + \nu_t \quad \nu_t \sim N(0, \mathrm{R})$$

Spead (innovation)

$$\nu_t = Y_t - F_t\theta_t$$

Kalman filter (state estimate)

$$\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + K_t\hat{\nu}_{t|t-1}$$

$$\boldsymbol{\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + K_t\left(Y_t - F_t\hat{\theta}_{t|t-1}\right)}$$

- $\hat{\theta}_{t|t-1}$: prior estimate
- $Y_t - F_t\hat{\theta}_{t|t-1}$: innovation
- K_t : weight on the innovation

Posterior error

$$e_t = \theta_t - \hat{\theta}_{t|t}$$

$$e_t = \theta_t - \hat{\theta}_{t|t-1} - K_t\left(Y_t - F_t\hat{\theta}_{t|t-1}\right)$$

$$e_t = \theta_t - \hat{\theta}_{t|t-1} - K_t\left(F_t\theta_t + \nu_t - F_t\hat{\theta}_{t|t-1}\right)$$

$$e_t = (\mathrm{I} - K_tF_t)\left(\theta_t - \hat{\theta}_{t|t-1}\right) - K_t\nu_t$$

$$e_t = (\mathrm{I} - K_tF_t)\left(\theta_t - \hat{\theta}_{t|t-1}\right) - K_t\nu_t$$

$$e_t = (\mathrm{I} - K_tF_t)e_t^- - K_t\nu_t$$

Posterior covariance

$$P_{t|t} = \mathrm{Var}(e_t)$$

$$P_{t|t} = \mathrm{Var}[(\mathrm{I} - K_tF_t)e_t^- - K_t\nu_t]$$

$$\boldsymbol{P_{t|t} = (I - K_tF_t)P_{t|t-1}(I - K_tF_t)^T + K_tRK_t^T}$$

Find weight (K_t) to minimize posterior covariance $P_{t|t}$

$$\frac{\partial P_{t|t}}{\partial K_t} = 2(I - K_t F_t) P_{t|t-1} F_t^T + 2K_t R = 0$$

$$-P_{t|t-1} F_t^T + K_t F_t P_{t|t-1} F_t^T + K_t R = 0$$

$$K_t \left(F_t P_{t|t-1} F_t^T + \mathrm{R} \right) = P_{t|t-1} F_t^T$$

$$K_t = P_{t|t-1} F_t^T \left(F_t P_{t|t-1} F_t^T + \mathrm{R} \right)^{-1}$$

Kalman Gain

$$\boldsymbol{K_t = P_{t|t-1} F_t^T \left(F_t P_{t|t-1} F_t^T + R \right)^{-1}}$$

State distribution

$$\theta_t \mid \mathrm{Y}_{1:t-1} \sim N(a_t, P_t)$$

$$a_t = \hat{\theta}_{t|t-1}$$

$$P_t = P_{t|t-1}$$

Conditional observation distribution

$$\mathrm{Y}_t = F_t \theta_t + \nu_t$$

$$\mathbb{E}[\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1}] = F_t \mathbb{E}[\theta_t \mid \mathrm{Y}_{1:t-1}] + \mathbb{E}[\nu_t]$$

$$\mathbb{E}[\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1}] = F_t \hat{\theta}_{t|t-1}$$

$$\hat{Y}_{t|t-1} = F_t a_t$$

$$\mathrm{Var}[\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1}] = \mathrm{Var}[\, F_t \theta_t + \nu_t \mid \mathrm{Y}_{1:t-1}]$$

$$\mathrm{Var}[\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1}] = \mathrm{Var}[F_t \theta_t \mid \mathrm{Y}_{1:t-1}] + \mathrm{Var}[\nu_t \mid \mathrm{Y}_{1:t-1}]$$

$$Var[Y_t \mid Y_{1:t-1}] = F_t P_t F_t^T + R = S_t$$

$$\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1} \sim N(F_t a_t, \mathrm{S}_t)$$

Spread (Innovation) distribution

$$\upsilon_t \mid Y_{1:t-1} = Y_t - F_t a_t$$

$$\upsilon_t \mid Y_{1:t-1} \sim N(0, \mathrm{S}_t)$$

Multivariate normal density

$$\mathbb{P}(\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1}) = \mathbb{P}(v_t \mid Y_{1:t-1}) = \frac{1}{\sqrt{2\pi \mathrm{S}_t}} \exp\left[-\frac{1}{2} v_{\mathrm{t}}^{\mathrm{T}} \mathrm{S}_{\mathrm{t}}^{-1} v_t\right]$$

Maximum Likelihood (define Q and R)

$$\log \mathrm{L} = \sum_{\mathrm{t}=1}^{\mathrm{T}} \log \mathbb{P}(\mathrm{Y}_t \mid \mathrm{Y}_{1:t-1})$$

$$\log \mathrm{L} = -\frac{1}{2} \sum_{\mathrm{t}=1}^{\mathrm{T}} [\log(2\pi) + \log(\mathrm{S}_t) + v_{\mathrm{t}}^{\mathrm{T}} \mathrm{S}_{\mathrm{t}}^{-1} v_t]$$

$$\boldsymbol{\log L = -\frac{1}{2} \sum_{t=1}^{T} \left[\log(S_t) + \frac{v_t^2}{S_t}\right]}$$

Where

$$P_t = P_{t|t-1} + \mathrm{Q}$$

$$\mathrm{S}_t = F_t P_t F_t^T + R$$